

Sprint 6

Completed the remaining core operational functionality of the Container Management System. The frontend now supports container management, approval processing and lifecycle tracking through an integrated React interface. The operational dashboard was enhanced with live statistics, analytics and Quick Scan functionality, providing users with real-time visibility of container status. During this sprint, the Observation module was removed to simplify the workflow and improve maintainability, ensuring the final application focused on the core lifecycle management requirements. By the end of the sprint, the complete container lifecycle had been successfully demonstrated through the integrated frontend and backend.

ID	User Story	Priority	Story Points	Status
PB22	Develop Container Management Interface	Must Have	5	Done
PB23	Implement Approval Workflow Interface	Must Have	8	Done
PB24	Enhance Dashboard with Operational Metrics	Must Have	5	DONE
PB25	Implement Quick Search	Should Have	5	Done
PB26	Display Container Details	Must Have	5	Done
PB27	Implement Recent Activity Panel	Should Have	3	Done
PB-028	Integrate Operational Analytics	Should Have	5	Done

• **Development Tasks • PB-22 – Container Management**

- Develop container list page
- Implement container detail view
- Develop movement workflow
- Display lifecycle information
- Implement role-based movement actions

• **PB-23 – Approval Workflow**

- Develop approvals page
- Display pending approvals
- Implement Supervisor approval review
- Reset lifecycle following approval
- Validate approval permissions
- Complete end-to-end approval workflow

• **PB-24 – Dashboard**

- Display operational summary cards
- Display pending approvals count
- Display production statistics
- Display cleaning statistics
- Add lifecycle warning information
- Prepare dashboard for reporting charts

GitHub Evidence

- **23 Jul 2026** – Complete end-to-end container lifecycle approval workflow on front end.
- **24 Jul 2026** – Improve container dashboard with Material UI chips.
- **24 Jul 2026** – Remove observation module and simplify lifecycle workflow.
- **25 Jul 2026** – Complete lifecycle system with analytics and quick scan.
- **03 Aug 2026** – Refine lifecycle expiry and Supervisor Reset workflow
- **03 Aug 2026** – Improve approval workflow and complete lifecycle validation
- **04 Aug 2026** – Complete dashboard analytics and regression endpoints
- **05 Aug 2026**—Final lifecycle testing and workflow refinements

Definition of Done

- Container management interface completed.
- Approval workflow interface operational.
- Lifecycle reset verified.
- Dashboard displays live operational statistics.
- Analytics integrated into the dashboard.
- Quick Scan implemented.
- Recent Activity panel operational.
- End-to-end workflow successfully tested.
- Code committed to GitHub.
- Documentation updated.

Sprint Review

Sprint 6 successfully completed the remaining functional requirements of the Container Management System. The React frontend was fully integrated with the backend workflow engine, enabling complete lifecycle management through a responsive user interface. Dashboard analytics, Quick Scan, approval processing, and operational reporting were completed, providing users with improved visibility of container activity.

During the sprint, significant improvements were made to the lifecycle engine. QA approval, Supervisor approval, lifecycle expiry, lifecycle reset, and production validation were refined and thoroughly tested to ensure compliance with the defined business rules. By the end of the sprint, the application supported the complete container lifecycle from initial receipt through repeated production cycles.

The application is now functionally complete and has entered the final testing, evaluation, and dissertation phase.

What Went Well

- Container management interface successfully completed.
- End-to-end approval workflow fully integrated.
- Dashboard analytics provided improved operational visibility.
- Quick Scan simplified container lookup.
- Simplifying the workflow by removing the Observation module improved maintainability.
- Frontend and backend integration remained stable throughout development.

Challenges

- Synchronising frontend updates with backend workflow logic.
- Debugging lifecycle state transitions.
- Maintaining consistent lifecycle reset behaviour after Supervisor approval.
- Refining the dashboard while preserving responsiveness.
- Balancing feature enhancements with project scope and deadlines.

Project Status

At the conclusion of Sprint 6: •

Backend workflow engine completed.

- React frontend completed.
- Authentication and role-based access control operational.
- Container lifecycle management completed.
- QA and Supervisor approval workflows completed.
- Dashboard and analytics implemented.
- Quick Scan implemented.
- Recent Activity implemented.
- End-to-end lifecycle validated.
- Final dissertation and supporting documentation remained in progress.